

Resume project sections

Verified wording - October 7, 2026

Humanoid Locomotion and Robustness

- Built an Isaac Lab-to-MuJoCo PPO evaluation pipeline; evaluated 3,600 simulation episodes across five randomization settings and three training seeds, with matched push-test falls decreasing from 341/360 to 13/360 under default randomization.
- Packaged a hash-verified replay suite for a trained 12-DoF biped; detected 40/40 injected sensor/action/scoring regressions and reproduced all five seeded nominal trajectories exactly on the same host.

MetaNavT - Hybrid Retrieval and Robotics Experiment Search

- Benchmarked neural retrieval on 57,638 FiQA documents and 648 test queries; BGE-small achieved nDCG@10 of 0.385 versus 0.237 for matched BM25, with paired bootstrap gain 0.148 [0.124, 0.174].
- Served neural retrieval over 600 robotics experiment files (5,633 nodes) through FastAPI/pgvector; measured 1.32 s p95 at 16 concurrent clients on an H100, with zero failures across 48 measured load-test requests.

Attribution and scope. Credit the original retrieval/data APIs to the six-person capstone contribution and these 2026 extensions separately. Humanoid results are simulation-only. HTTP p95 at 16 clients has 16 samples, with 48 total measured requests across 1/4/16 clients. Full methods and alternate role-specific bullets are in resume-projects.txt and revamp-report.pdf.